



Ziyan Huang

+86-13850318831

mathziyanhuang@163.com

CET-6 Score: 502

EDUCATION

•Hohai University (211 Program)

Sep 2020 – Jun 2024

B.S. in Information and Computing Science; GPA: 4.29/5.0

Nanjing, China

•Central South University (985 Program)

Sep 2024 – Jun 2027 (Expected)

M.S. in Mathematics; GPA: 3.59/4.0

Changsha, China

Supervisor: Hongqiao Wang, Associate Professor

PUBLICATION

•Physen-N2N: A Physics-Guide Self-Supervised Learning for Nonblind Image Deconvolution in Biased Complex Noise (Under Review)

First author

Design the algorithm, conduct experiments, and author the paper

- **Self-Supervised Learning** This study presents a self-supervised deep learning framework designed to solve inverse problems under biased complex noise conditions. The framework first maps noisy data to a clean image combined with the noise expectation. It then incorporates a physical model into the loss function as an auto-regressive training strategy, enabling joint optimization of both the clean image and the noise distribution expectations.
- **Theoretical Analysis** A theoretical framework is proposed to elucidate the proposed PN2N method based on joint optimization through natural constraints of physical models, revealing the characteristic natural constraint contained within blur kernel.

•Physics Law Guided Self-Supervised Learning Bayesian Framework for Computational Imaging with Reliability (Under Review)

First author

Conduct experiments, and author the paper

- **Self-Supervised Learning** A self-supervised learning framework for computational imaging that integrates physical forward models into the loss function, combining physical modeling with deep learning to address imaging blur caused by optical system imperfections.
- **Uncertainty Qualification** Through well-trained neural network sampling via self-supervised learning, the pCN algorithm is employed to iteratively optimize and update the uncertainty in the identified reconstruction regions, contributing to the quantification of reconstructed image credibility in this field.

RESEARCH EXPERIENCE

•Empathetic AI: Enhancing Empathy and Multidimensional Understanding in LLMs

Nov 2024 – Feb 2025

Peking University Changsha Institute for Computing and Digital Economy

LLM Algorithm Intern

- **Data Processing and Generation** Web scraping is conducted on quasi-psychological counseling QA data and text classification data from platforms such as Weibo and Douban. The data is cleaned, filtered, and formatted to construct high-quality QA pair datasets for LoRA fine-tuning and labeled datasets for text classification. A prompt engineering workflow is designed, leveraging DeepSeek-V3 to diversify and enhance QA rewrites, thereby improving corpus diversity. Based on testing feedback, specific sentence structures and domains are optimized iteratively to continuously enhance the model's performance
- **Dialogue Classification and Implementation** Multi-task training is performed using the BERT model on text data with emotion and event labels, enabling the AI to identify user emotion categories and event types. Graded alerts are implemented based on emotional intensity and event attributes.
- **Empathy Capability Optimization** An empathy quantification evaluation model is researched and introduced as the core reward criterion for reinforcement learning. A multi-dimensional reward function is constructed by combining text fluency and emotional alignment.

TECHNICAL SKILLS AND INTERESTS

Research Interests: Computational Imaging, LLM, Deep Learning, Uncertainty Quantification, 3D Reconstruction

Skills and Strengths:

- Proficient in DL, with project experience in downstream tasks and fine-tuning of LLM
- Experience in Optical: Solid understanding of imaging principles and optical theory
- Familiar with Bayesian Neural Networks and Gaussian Process Regression for statistical learning applications

HONORS AND AWARDS

- Honorable Mention (MCM) • First-Class Scholarship (CSU) • Second-Class Scholarship (CSU)
- Academic Excellence Scholarship (HHU) • Science and Technology Innovation Scholarship (HHU)